

REACT JS

LECTURE 1

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AGENDA

- Introduction about React
- Single page application
- Virtual Dom
- Environment setup
- React app structure
- What is JSX ?
- Building Reusable components
 - Handling events
- Component lifecycle
- State and Props.

an open-source JavaScript library that is used for building user interfaces specifically for single-page applications.

React first deployed on Facebook's newsfeed in 2011 and on Instagram.com in 2012 , 2013(open source).

component based and state driven.

React allows developers to create large web applications that can change data, without reloading the page. The main purpose of React is to be fast, scalable, and simple

Last version :19.2.0

WHY REACT

1- High Performance

ReactJS main feature i.e

Virtual DOM that results in high performance of the app

2- Easy to learn.

3- Easy Transition to react native.

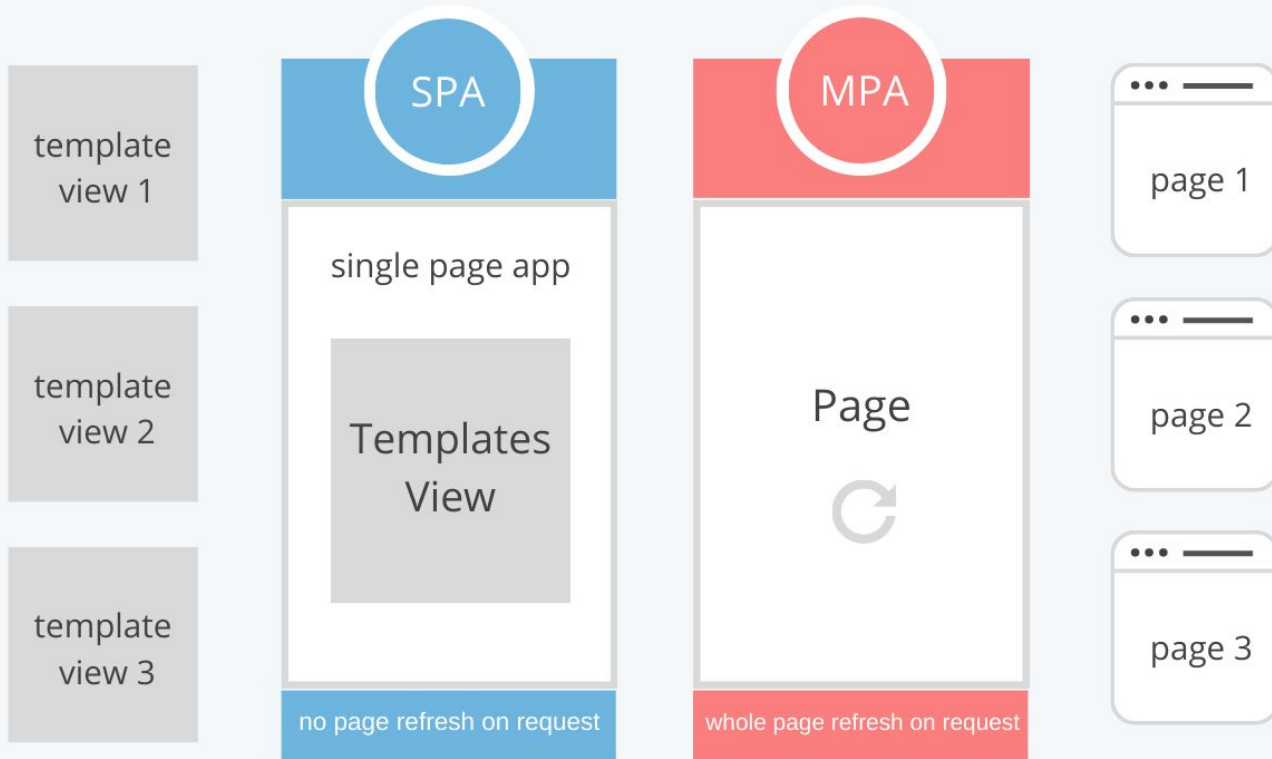
4- Component Style Architecture.

5- Reusable Component.

SINGLE PAGE APPLICATION

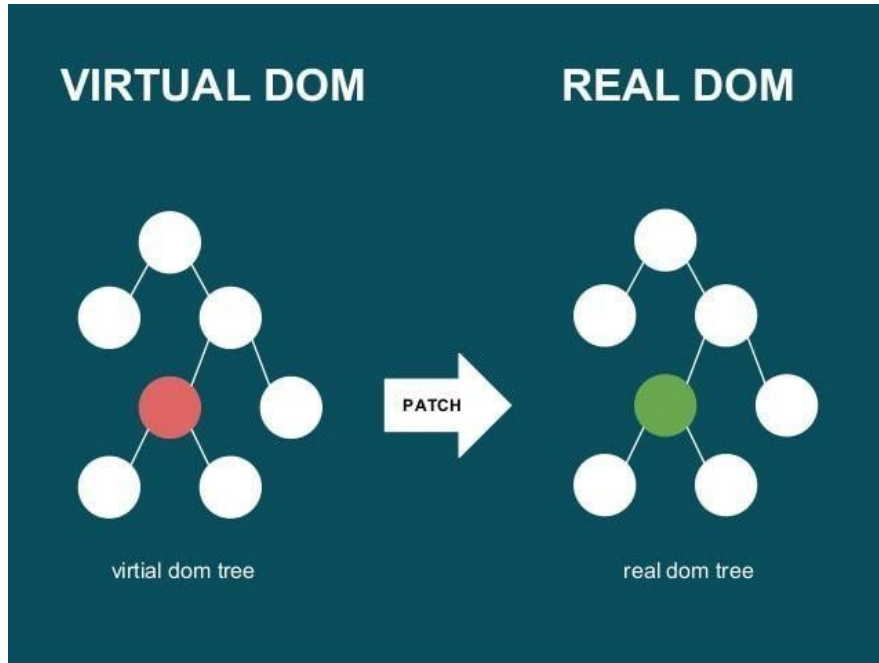
A Single Page Application (SPA) is a web application or website that interacts with the user by dynamically rewriting the current page rather than loading entire new pages from the server. This approach provides a more fluid and responsive user experience, similar to that of a desktop application.

SINGLE PAGE APPLICATION

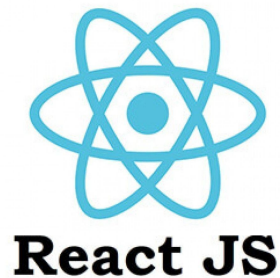


VIRTUAL DOM

The virtual DOM (VDOM) is a programming concept where an ideal, or “virtual”, representation of a UI is kept in memory and synced with the “real” DOM by a library such as ReactDOM. whenever there is a change in state of any element, a new Virtual DOM tree is created. This new Virtual DOM tree is then compared with the previous Virtual DOM tree and make a note of the changes. After this, it finds the best possible ways to make these changes to the real DOM. Now only the updated elements will get rendered on the page again.



GETTING STARTED



1-first : you have to install node js

2-

deprecated :

`npx create-react-app appName`

`npm start (to run the project)`

or using vite (recommended to use)

`npm create vite@latest`

`npm run dev (to run the project)`

CREATE-REACT-APP VS VITE

CREATE-REACT-APP

- Complete “starter kit” for React applications
- Everything is already configured: Prettier, Jest, etc.
- Uses slow and outdated technologies (i.e. webpack)
- ✅ Use for tutorials or experiments
- ❌ Don't use for a real-world app

VITE

- Modern build tool that contains a template for setting up React applications (esBuild)
- Need to manually set up Jest (and others)
- Extremely fast hot module replacement(HMR) and bundling
- ✅ Use for modern real-world apps

CREATE-REACT-APP VS VITE

Aspect	Create React App (CRA)	Vite
Command	<code>npx create-react-app</code>	<code>npm create vite@latest --template react</code>
Build Tool	Webpack + Babel	ESBuild (dev) + Rollup (prod)
Performance	Slower build and HMR	Faster build and HMR
Configuration	Requires ejecting for config access	Simple and extensible config
Dependency Handling	Centralized, can be slow	ESM-based, faster
Popularity	Established but declining	Growing rapidly
Development Experience	Heavier, slower dev server	Lightweight, fast dev server

WHAT IS JSX

Stands for javascript xml. JSX is an XML/HTML-like syntax used by React that extends ECMAScript so that XML/HTML-like text can co-exist with JavaScript/React code.

- Declarative syntax to describe what components look like and how they work.
- Components must return a block of JSX.
- Extension of JavaScript that allows us to embed JavaScript, CSS, and React components into HTML

Without JSX : `React.createElement("a",{href:"#"},"about")`

With JSX : `<a href="#">about</a>`

RULES OF JSX

1. Return a single root element

To return multiple elements from a component, wrap them with a single parent tag.

2. Close all the tags

JSX requires tags to be explicitly closed: self-closing tags like `<img>` must become `<img />`, and wrapping tags like `<li>oranges` must be written as `<li>oranges</li>`

RULES OF JSX

3. camelCase most of the things!

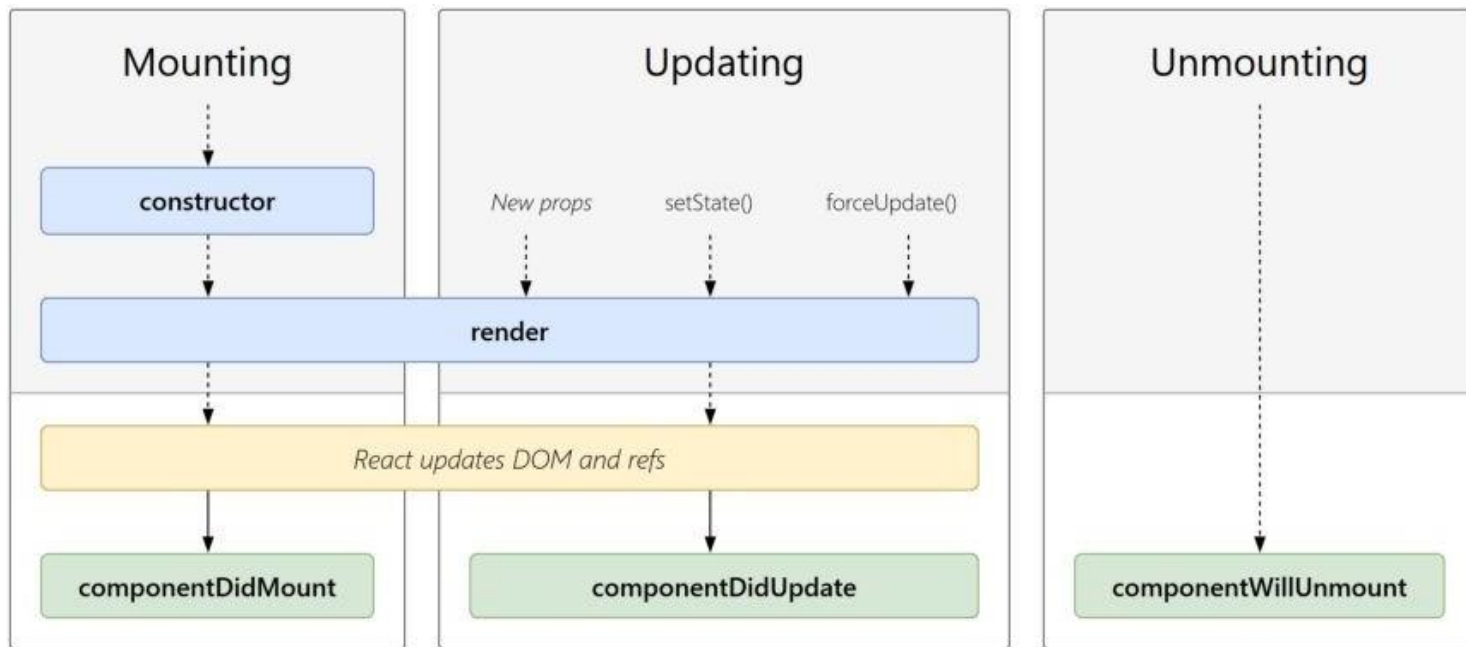
JSX turns into JavaScript and attributes written in JSX become keys of JavaScript objects. In your own components, you will often want to read those attributes into variables. But JavaScript has limitations on variable names. For example, their names can't contain dashes or be reserved words like `class`.

COMPONENT LIFECYCLE

Each component goes through three phases:

1. Mounting
2. updating
3. unmounting

COMPONENT LIFECYCLE (CLASS COMPONENT)

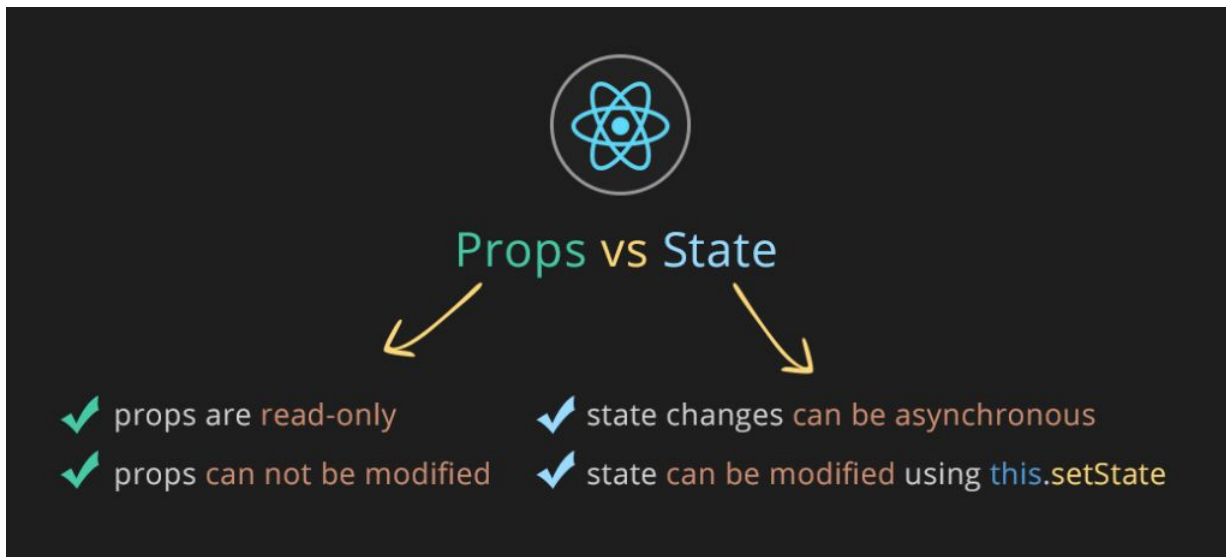


PROPS

1. Props are used to pass data from a parent component to a child component, while state is used to manage data within a component.
2. Props are immutable and cannot be changed within a component, while state is mutable and can be updated using the `setState` function.
3. Props are passed down from the parent component and are read-only within the child component, while state is only accessible within the component where it is defined.
4. Props can be used to customize the behavior or appearance of a component, while state is used to keep track of information that can change over time.

STATE

The state is a built-in React object that is used to contain data or information about the component. A component's state can change over time; whenever it changes, the component re-renders.



HANDLING EVENTS

```
<button onClick={adduser}>Click Me</button>
```

```
<button onClick={()=>this.handleClick()}>  
Click Me</button>
```

THANK YOU